Search Learning Map

General problem solving -> search formulation -> execution -> evaluation

DEFINE	>	PLAN	>	TRY	>	ITERATE
What is the problem? What counts as good?		Compare approaches Choose strategy		Execute algorithm Track state		Correct? Complete? Optimal? Efficient?

Closed loop: Iterate can send you back to Try, Plan, or Define.

Inside DEFINE - Search Problem Formulation

Initial State + Actions + Transition Model + Goal Test + Path Cost
Also define the objective: any solution? fewest edges? minimum weighted cost? efficiency constraints?

Inside PLAN - Choose the Approach

First: is search the right family? Alternatives may include optimization, CSP, probability, ML, RL, etc.
If search fits: compare BFS, UCS, A*, and other strategies against the objective.

Inside TRY - Search Algorithm Execution

Frontier + Selection Rule + State Tracking + Duplicate Handling + Termination + Path Reconstruction

Inside ITERATE - Close the Loop

Correct?   Complete?   Optimal for the objective?   Efficient enough (time / space / expansions)?

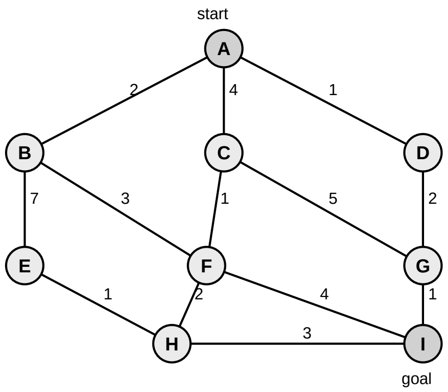
BFS - Exact Execution Lifecycle

enqueue = put into queue; dequeue = remove from front; visited = already discovered; expanded = neighbors examined.
0. Initialize: frontier=[start], visited={start}, parent[start]=None
1. Select: dequeue first node (FIFO). 2. Goal test: if current==goal, reconstruct and return.
3. Expand: examine every neighbor.
4. New child: visited.add(child) -> parent[child]=current -> enqueue(child).
5. Repeat while frontier nonempty. Frontier empty before goal -> no path.

Timing: ENTER frontier => visited + parent + enqueue. LEAVE frontier => goal test, then expand.
Reconstruct: goal -> parent -> ... -> start, then reverse.

BFS Manual Trace - Guided Practice

Start=A, Goal=I. Undirected. Ignore weights for BFS. Process neighbors alphabetically.



Step	Frontier BEFORE	Deq	Goal?	New discoveries	Frontier AFTER	Parent updates
0						
1						
2						
3						
4						
5						
6						
7						
8						

Returned path: ______________________________

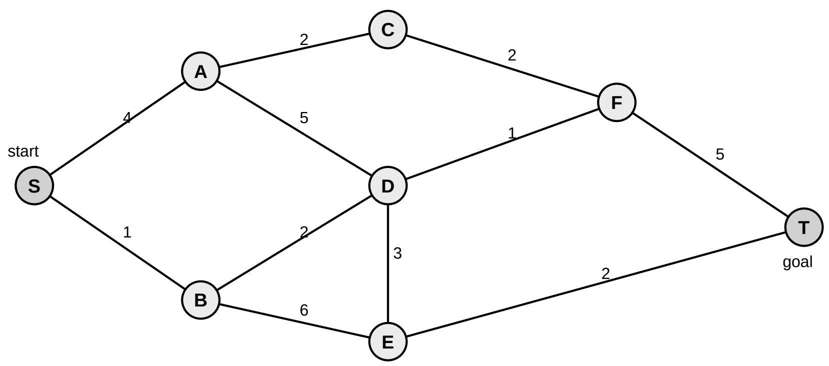
Backtrack goal -> start: ______________________________

Expansion order: ______________________________

Why optimal? ______________________________

Independent BFS Test + Mastery Gate

Do this page without help. Start=S, Goal=T. Ignore weights for BFS. Process neighbors alphabetically.



Step	Frontier BEFORE	Deq	Goal?	New discoveries	Frontier AFTER	Parent updates
0						
1						
2						
3						
4						
5						
6						
7						

BFS Mastery Gate

[ ] enqueue vs dequeue is automatic
[ ] start: frontier + visited + parent=None
[ ] new child: visited + parent + enqueue
[ ] revisited child: skip; parent unchanged
[ ] frontier correct after dequeue
[ ] reconstruct goal -> start, then reverse
[ ] BFS = fewest edges, not weighted cost
[ ] node-finished vs search-finished is clear

Decision: <=1 minor slip -> UCS. 2+ timing/state errors -> one more BFS trace.

Derive UCS from BFS, then A* from UCS

BFS -> UCS: objective? frontier/priority/state changes? why is first discovery no longer enough?
UCS -> A*: what does h(n) add? If h(n)=0, what should A* reduce to?